

YU-WEN CHEN

MACHINE LEARNING RESEARCHER - PHYSICAL AI & SATELLITE REMOTE SENSING

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PROFESSIONAL SUMMARY

Machine learning researcher working on physics-ML hybrid models for satellite remote sensing. NASA FINESST Graduate Fellow; Ph.D. expected Dec 2026. The work combines 3D radiative-transfer physics with deep learning, on PyTorch pipelines that run over **3M+ satellite frames** on HPC (SLURM). Main methods are inverse problems and calibrated uncertainty quantification with conformal prediction, applied to large climate and remote-sensing datasets.

CORE COMPETENCIES

ML / Physics-Hybrid Modeling. PyTorch, physics-ML hybrid methods (3D radiative transfer + deep learning), physics-informed neural networks, spatiotemporal modeling, gradient-boosted trees, deep ensembles, tabular deep learning

Evaluation & Uncertainty. Conformal prediction, calibrated uncertainty quantification, date-blocked k-fold cross-validation, inverse problems, Bayesian optimization, statistical inference

Climate & Remote-Sensing Data. Large-scale satellite remote sensing (OCO-2, MODIS, TEMPO, PREFIRE), shortwave and longwave radiative transfer (EaR3T, libRadtran), photon path-length analysis, multi-sensor collocation, geospatial search

Languages & Infra. Python (expert), PyTorch, Fortran, AWS, HPC/SLURM, Linux, Git

EXPERIENCE

University of Colorado Boulder Jan 2022 - Present
NASA FINESST Graduate Fellow / Graduate Researcher

- Cut CO2 retrieval error by **76%** in cloud-adjacent scenes by building a physics-ML hybrid correction (3D radiative transfer + deep learning); peer-reviewed in *Atmos. Meas. Tech.* (2025).
- Developed an ensemble ML system (XGBoost, Gaussian-NLL deep ensembles, TabM) with calibrated conformal-prediction uncertainty to detect and remove satellite retrieval anomalies, validated via date-blocked k-fold cross-validation.
- Engineered a large-scale climate data pipeline collocating OCO-2/MODIS satellite feeds (banded KD-Tree geospatial search, Numba-accelerated on HPC/SLURM, **3M+ frames**); scaled the EaR3T prototype to production OCO-2 processing using agentic AI tooling (Claude, GitHub Copilot).

University of Colorado Boulder Aug 2024 - Present
Graduate Researcher - Arctic Surface Properties

- Simulated mid- and far-infrared radiances for **PREFIRE** satellite channels and broadband cloud radiative effect with libRadtran, building the atmospheric and cloud profile inputs from MODIS and climatology data. Open source (GPL-3.0, ~800 commits) at github.com/ywchen-tw/arcsix_sfc.

Academia Sinica - Research Center for Environmental Changes Jan 2019 - Jul 2021
Research Assistant - Pollution Transport Analysis

- Built autoregressive spatiotemporal generative models (CNN/LSTM) to predict station-level ozone evolution in complex terrain; presented at the AI: Modeling Oceans and Climate Change (AIMOCC 2021) Workshop at ICLR 2021.

EDUCATION

Ph.D., Atmospheric & Oceanic Sciences - University of Colorado Boulder (NASA FINESST Fellow) Expected Dec 2026

M.S., Atmospheric & Oceanic Sciences - University of Colorado Boulder 2023

M.S., Chemistry - National Taiwan University 2018

B.S., Chemistry (Minor in Atmospheric Science) - National Taiwan University 2016

SELECTED HONORS

- NASA FINESST Award (13% selection rate) - University of Colorado Boulder, 2024
- Summer School for Inverse Modeling of Greenhouse Gases - Colorado State University, 2024
- Government Scholarship to Study Abroad - Taiwan Ministry of Education, 2023